

FEDERATED UNLEARNING THESIS

Month 2-3 Progress Report

Gate 2 closure (hand-written FedAvg) and Month 3 Week 9 (IID vs. Non-IID)

Generated 2026-09-09 from saved metrics.json files under results/. No number below is hand-typed.

Gate 2 — FedAvg from Scratch (closed 2026-09-07)

Federated Averaging was implemented by hand and verified against 13 synthetic tests before being run on real MNIST data. The canonical protocol uses 5 IID clients, full participation, one local epoch per round, batch size 128, and five communication rounds, matched against a centralized SGD baseline trained on the same 51,000-example split from the same initial weights.

Canonical result and controlled variants

Setting	Centralized acc.	FedAvg acc.	FedAvg F1	Gap (pp)
Canonical (K=5, E=1)	94.76%	90.99%	90.86%	-3.77
K=10 (double clients)	94.76%	89.55%	89.37%	-5.21
E=2 (double local epochs)	96.17%	92.31%	92.21%	-3.86

Every predicted number the student wrote before either variant ran (examples per client, optimizer-step totals, exposures, communication bytes) matched the executed result exactly. Doubling client count widened FedAvg's gap to centralized at double the communication cost; doubling local epochs raised both methods' absolute accuracy without changing communication or the relative gap. Reviewed and passed 2026-09-07.

Month 3, Week 9 — IID vs. Non-IID FedAvg

With Gate 2 closed, Month 3 asks what happens to FedAvg once clients stop being interchangeable random samples. A pathological shard-based Non-IID partition (McMahan et al. 2017, Section 3: sort by label, two shards per client) was compared against the existing IID partition, training from identical initial weights under the same K=5, C=1, E=1, B=128, R=5 protocol as the Gate 2 canonical run.

Partition	Test accuracy	Test macro F1
IID	90.99%	90.86%
Non-IID (2 shards/client)	64.01%	57.80%

The IID number exactly reproduces Gate 2's canonical FedAvg result — a built-in sanity check that this new runner is wired correctly, not a new measurement. The Non-IID run drops 26.98 percentage points. Each Non-IID client's saved class histogram is dominated by one or two digits, while every IID client stays close to flat across all ten — the heterogeneity is visible in the saved data, not just asserted. This is one Non-IID severity, not a Dirichlet sweep, and not a FedProx comparison; those are Weeks 10 and 11.

Client class distributions: IID vs. Non-IID (Month 3, Week 9)

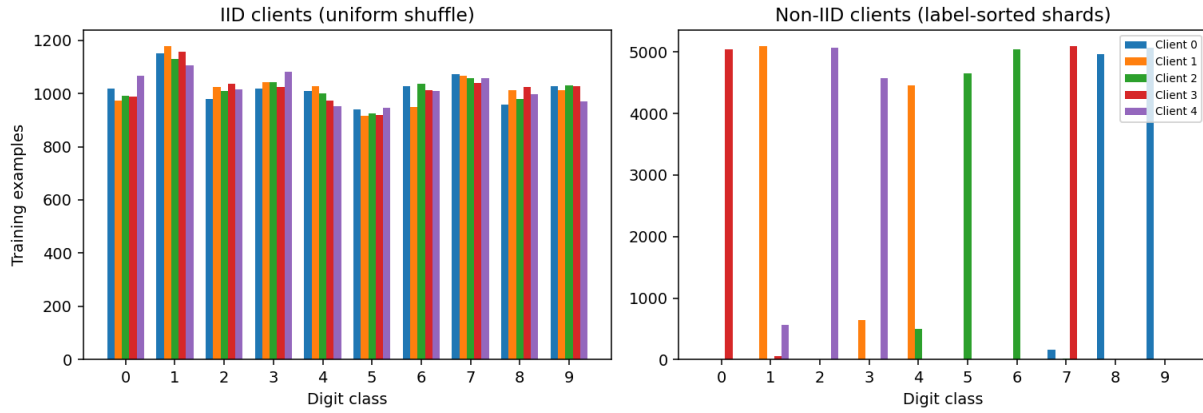


Figure: per-client training-set class distribution, IID (left) vs. Non-IID (right).

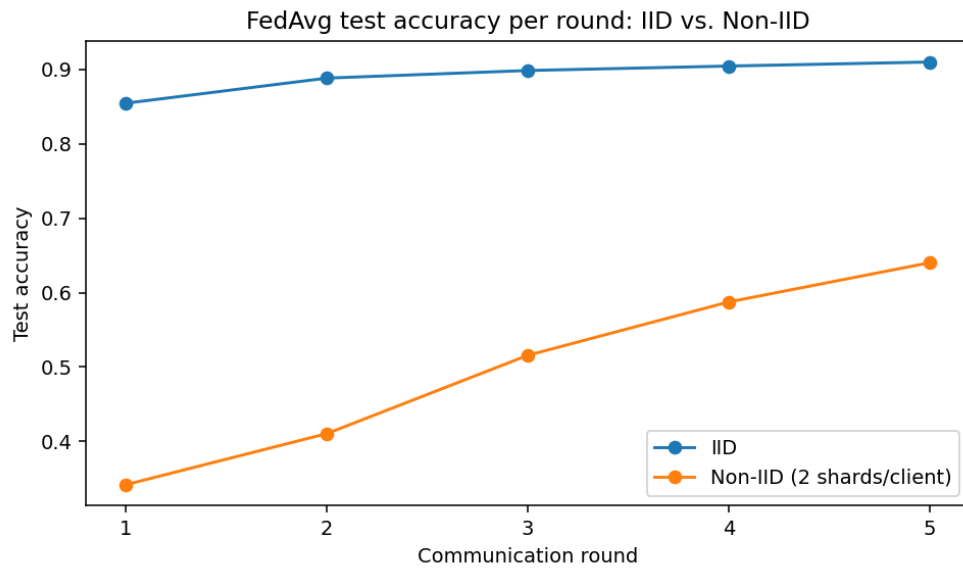


Figure: FedAvg test accuracy per communication round, IID vs. Non-IID.

Month 3, Week 10 — Dirichlet(α) Severity Gradient

Week 9 showed one pathological extreme. Week 10 asks how FedAvg degrades as heterogeneity is dialed continuously, using Hsu, Qi & Brown (2019)'s per-class Dirichlet(α) label-skew scheme: a smaller α concentrates each digit class onto fewer clients; a larger α keeps classes close to uniformly spread. Three configs ($\alpha = 1.0, 0.5, 0.1$) each retrain centralized SGD and hand-written FedAvg from identical initial weights under the same protocol as Week 8/9.

Setting	Centralized	FedAvg	Gap (pp)
IID (Week 8/9)	94.76%	90.99%	-3.77
Dirichlet $\alpha=1.0$	94.76%	90.59%	-4.17
Dirichlet $\alpha=0.5$	94.76%	89.83%	-4.93
Dirichlet $\alpha=0.1$	94.76%	71.60%	-23.16
Pathological shards (Week 9)	94.76%	64.01%	-30.75

The severity gradient is nonlinear in α , matching Hsu et al.'s own finding: $\alpha=1.0$ and $\alpha=0.5$ both stay within about 1.4 points of the IID result, while $\alpha=0.1$ causes a much larger drop. Dirichlet skew also unevenly skews client *quantity*, not just label mix — training-set sizes ranged from about 6,100 to 16,700 examples across the 5 clients at every α tested, a real, literature-typical side effect of the per-class draw. Placing Week 9's pathological-shard result in the same table shows “Non-IID” is a spectrum: Dirichlet $\alpha=0.1$ is meaningfully less severe than the 2-shard scheme, even though both are informally called “severe.”

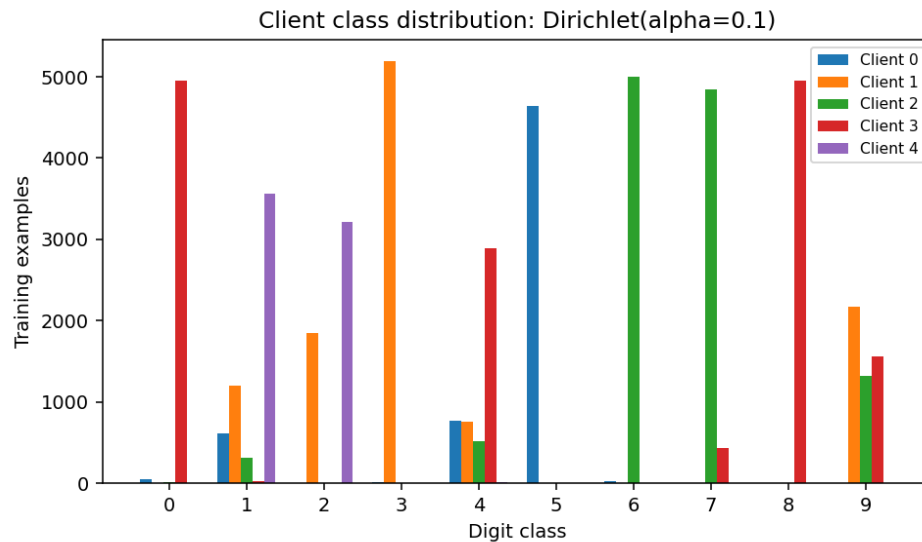


Figure: per-client training-set class distribution at Dirichlet $\alpha=0.1$.

Month 3, Week 11 — Does FedProx Help?

Li et al. (2020)'s FedProx adds a proximal term $(\mu/2) \cdot \|w - w_{\text{global}}\|^2$ to each client's local loss, specifically to limit the client drift Weeks 9-10 measured. $\mu=0$ is verified by test to reproduce hand-written FedAvg exactly (client weights and a full server round), so FedProx here is a strict generalization, not a parallel reimplementation. The experiment reuses Week 10's exact Dirichlet($\alpha=0.1$) partition, seeds, and initial weights, sweeping $\mu \in \{0.01, 0.1, 1.0\}$ at two local-epoch counts.

Method	E=1 accuracy	E=5 accuracy
FedAvg ($\mu=0$)	71.60%	76.28%
FedProx ($\mu=0.01$)	71.43%	75.33%
FedProx ($\mu=0.1$)	69.65%	69.65%
FedProx ($\mu=1.0$)	60.15%	57.30%

FedProx did not beat FedAvg at any swept μ , in either local-epoch setting — reported plainly, per the plan's own rule that a negative result must remain reportable. This does not contradict Li et al.'s paper: their clearest reported gains come under systems heterogeneity (clients doing genuinely different amounts of local work) and longer training horizons, neither of which this 5-round, uniform-local-epoch experiment tests. Three checks rule out an implementation bug: $\mu=0$ equivalence is tested, not assumed; the E=1 and E=5 round-by-round histories genuinely differ; and one striking coincidence — $\mu=0.1$ landing on the identical 69.65% at both E=1 and E=5 — was traced to the underlying per-round trajectories and confirmed real, not a copy-paste error. Full discussion: [reports/month3_week11_fedprox.md](#).

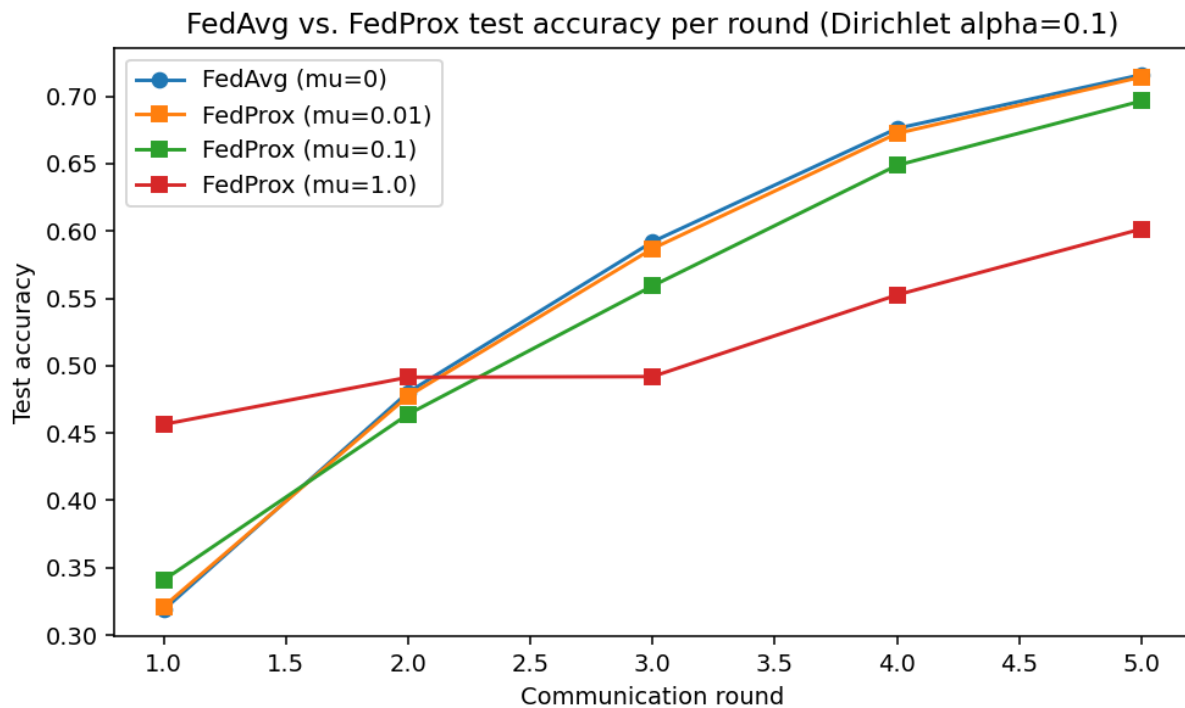


Figure: FedAvg vs. FedProx test accuracy per round at Dirichlet $\alpha=0.1$, E=1.

Next: completing Month 3 (Gate 3)

Week 12 benchmarks FedAvg against FedProx across IID and Non-IID settings, which closes Gate 3. Federated Unlearning remains blocked until Gates 3 and 4 both close.